Alessio Cocchieri

NLP PhD Researcher · UniboNLP · University of Bologna

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About Me

Last-year NLP PhD candidate (ending Nov 2026) at UniboNLP, with 9+ publications at ACL, EMNLP, NAACL, and EACL. I specialize in LLM evaluation and knowledge distillation for low-resource NLP, with applications to high-stakes domains like medicine. Active reviewer for ARR and top-tier ML conferences like NeurIPS.

Education

PhD **University of Bologna**, Natural Language Processing — Bologna, Italy, Nov 2023 – present
- Focus: LLMs, RAG, Information Extraction, Benchmarking, Low-resource NLP
- Supervisor: Prof. Gianluca Moro — Research Group: UniboNLP

MSc **University of Bologna**, Artificial Intelligence — Bologna, Italy, Sept 2021 – Dec 2023
- Grade: 110/110 with honors

BSc **University of Bologna**, Computer Science — Bologna, Italy, Sept 2018 – July 2021
- Grade: 110/110 with honors

Experience

IBM Research, Research Scientist Internship (PhD Intern) — Dublin, Ireland
Natural Language Processing — LLM Factuality — Mar 2026 – May 2026, 3 months
- Created a novel benchmark for multimodal scientific fact-checking, targeting LLM long-form generation
- Coordinated a multi-team human annotation pipeline, overseeing annotation guidelines, quality control, and IAA
- Conducted systematic analysis exposing critical factuality fragilities in multimodal LLMs across scientific domains

IBM Research, Research Scientist Internship (Master Thesis) — Dublin, Ireland
Natural Language Processing — Named Entity Recognition — Mar 2023 – May 2023, 3 months
- Leveraged LLM distillation to improve smaller-sized models for zero-shot NER
- Produced two peer-reviewed publications — OpenBioNER (NAACL 2025) and ZeroNER (ACL 2025)
- Contributed to the IBM zshot library for zero-shot NER model inference

Publications

LLMs (Almost) Never Abstain Under Medical Uncertainty — Jan 2026
We introduce MedQAbstain, a benchmark for medical abstention under uncertainty, revealing that state-of-the-art LLMs systematically overcommit, rarely abstaining even when the question itself is hidden.
Alessio Cocchieri, et al.
(ACL 2026 (Main))

ReMedQA: Are We Done With Medical Multiple-Choice Benchmarks? — Jan 2026
We show that high LLM accuracy in medical MCQA masks severe inconsistency. We propose novel metrics to evaluate true reliability across MCQA formats.
Alessio Cocchieri, et al.
(EACL 2026 (Main))

What do you call a dog that is incontrovertibly true? Dogma: Testing LLM Generalization through Humor Jan 2025
We introduce Phunny, a novel benchmark using uncontaminated English puns, revealing that LLMs struggle with generalization even on simple tasks, consistently underperforming the human baseline.
Alessio Cocchieri, et al.
(ACL 2025 (Main))

Can Large Language Models Win the International Mathematical Games? Jan 2025
We introduce MathGames, a novel multimodal benchmark of age-graded math problems from an international competition, showing that frontier LLMs underperform compared to humans, including 11-year-olds.
Alessio Cocchieri, et al.
(EMNLP 2025 (Main))

OpenBioNER: Lightweight Open-Domain Biomedical NER Through Entity Type Description Jan 2025
We introduce a 110M BERT model that leverages descriptions for zero-shot Biomedical NER, outperforming GPT-4o, specialized LLMs, and GLiNER by up to 10% F1.
Alessio Cocchieri, et al.
(NAACL 2025 (Findings))

To Generate or to Retrieve? On the Effectiveness of Artificial Contexts for Medical Open-Domain Question Answering Jan 2024
We introduce MedGENIE, the first generate-then-read framework for open-domain medical QA, demonstrating the effectiveness of generated over retrieved contexts and significantly improving LLM RAG performance.
Giacomo Frisoni, *Alessio Cocchieri*, et al.
(ACL 2024 (Main))